

Project Task Plan

Project Title: Labor Market Trends and Wage Dynamics in North Carolina (2015–2024)

1. Description of the Data

The dataset used for this project focuses on labor market trends in North Carolina from 2015 to 2024. The data contains **120 observations**, representing monthly economic indicators across a ten-year period. The purpose of using this dataset is to analyze how employment levels, unemployment rates, and wage growth interact over time.

The dataset includes the following main variables:

- **Date:** Monthly time reference for each observation.
- **Year and Month:** Time identifiers used to analyze trends and seasonality.
- **State:** The geographic focus of the dataset, which is North Carolina.
- **Unemployment Rate:** The percentage of the labor force that is unemployed during each month.
- **Average Hourly Wage:** The estimated average hourly wage for workers in the labor market.
- **Employment (Thousands):** The total number of employed individuals measured in thousands.

The dataset represents economic indicators commonly tracked by institutions such as the **U.S. Bureau of Labor Statistics (BLS)** and similar public economic databases. These indicators are widely used to evaluate labor market conditions, economic growth, and workforce stability.

One limitation of the dataset is that the data is aggregated at the state level, which means it does not capture more detailed variations between cities or industries. However, the dataset still provides strong insight into **overall labor market trends**, making it appropriate for economic analysis and visualization using Tableau.

2. Research Questions

The goal of this project is to understand how key labor market variables interact and change over time. The analysis will focus on identifying patterns between employment, unemployment, and wage growth.

The three main research questions for the project are:

1. How has the unemployment rate in North Carolina changed from 2015 to 2024, and what overall trends can be observed in the labor market during this period?

2. What relationship exists between employment levels and average hourly wages in North Carolina, and can increases in employment help explain wage growth?

3. Are changes in unemployment associated with noticeable changes in employment levels and wages, particularly during periods of economic disruption or recovery?

These questions aim to identify patterns within the labor market and explore whether certain economic indicators move together or influence one another over time.

3. Pathways and Sub-Questions for the Analysis

To answer the main research questions, the analysis will follow several structured steps using data visualization and regression analysis.

First, the dataset will be explored through **descriptive analysis**. Line charts will be created in Tableau to visualize changes in unemployment rates, employment levels, and wages across the ten-year period. This step will help identify long-term trends and potential turning points in the data.

Second, the project will examine the relationship between variables using **scatter plots and trend lines**. For example, employment levels will be compared with wage levels to determine whether higher employment is associated with higher wages.

Third, a **regression analysis** will be conducted to evaluate whether certain variables can predict changes in others. For example, regression can help determine whether unemployment rates significantly affect wage levels or employment growth.

Additional guiding questions include:

- Do wages tend to increase during periods of higher employment?
- Are unemployment spikes followed by slower wage growth?
- Are there noticeable economic patterns across the 2015–2024 time period?

By combining visualization with statistical analysis, the project will aim to produce a deeper understanding of labor market dynamics in North Carolina.

- Renato Herran.